```
class test1:
  _b=None
  __c=None
  def __init__(self):
    self.a=101
    self._b=201
    self.__c=301

  def show1(self):
    print("a=",self.a)
    print("b=",self._b)
    print("c=",self.__c)

class test2(test1):
  def show2(self):
    pass
    print("a=",self.a)
    print("b=",self._b)
    #print("c=",self.__c) # This will show error as it is a strongly private variable
class test3(test2):
  def show3(self):
    pass
    print("a=",self.a)
    print("b=",self._b)
    #print("c=",self.__c) # This will show error as it is a strongly private variable

t1=test1()
t1.show1()
t2=test2()
t2.show2()
t3=test3()
t3.show3()
```

```
a= 101
b= 201
c= 301
a= 101
b= 201
a= 101
b= 201
```

Logical questions:

```
# Find the second largest element
a=[7,3,9,2,8]
a.sort()
print(a)
```

```
[2, 3, 7, 8, 9]
```

```
print("Second largest element: ", a[-2])
```

```
Second largest element:  8
```

```
# Move zeroes to the end
a=[0,1,0,3,12]
a.sort()
print(a)
```

```
[0, 0, 1, 3, 12]
```

```
b=len(a)
for i in range(b):
  if(a[i]==0): # //
    a[i]=a[i-b]

print(a)
```

```
[0, 0, 1, 3, 12]
```

```
a=[0,1,0,3,12]
for i in a:
  if(i==0):
    a.remove(i)
    a.append(i)

print(a)
```

```
[1, 3, 12, 0, 0]
```

```
# Find common elements of three sets
a=[1,2,3]
b=[2,3,4]
c=[3,4,5]
L=[]
for i in a:
  for j in b:
    for k in c:
      if(i==j and j==k):
        L.append(i)
print(L)
```

```
[3]
```

```
# OR
L1=[]
for i in a:
  if (i in b and i in c):
    L1.append(i)
print(L1)
```

```
[3]
```

```
# Find number of max consecutive ones
q=[1,1,0,1,1,1,0,1,1,1,1]
count=0
for i in q:
  if(i==1):
    count=count+1
  else:
    count=0

print(count) #SA
```

```
4
```

```
##
a=[1,2,3,4,5,6,7,8,9]
a[::2]=10,20,30,40,50

print(a)
##
a=[1,2,3,4,5,6,7,8,9]
a[::2]=10,20,30,40,50,60 # 60 is going out of len(a), toh not possible to assign it an index
print(a)
```

```
[10, 2, 20, 4, 30, 6, 40, 8, 50]
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
/tmp/ipykernel_2506/687651145.py in <cell line: 0>()
      6 ##
      7 a=[1,2,3,4,5,6,7,8,9]
----> 8 a[::2]=10,20,30,40,50,60
      9 print(a)

ValueError: attempt to assign sequence of size 6 to extended slice of size 5
```

```
a=[1,2,3,4,5]
print(a[3:0:-1])
```

```
[4, 3, 2]
```

```
def f(i, values=[]): # Here, values is a default list
  values.append(i)
```

```
  print(values)

f(1)
f(2)
f(3)
```

```
[1]
[1, 2]
[1, 2, 3]
```

```
fl1=['Apple','Berry','Cherry','Papaya']
fl2=fl1
fl3=fl1[:]
fl2[0]='Guava'
fl3[1]='Kiwi'

sum=0
for ls in (fl1,fl2,fl3):
  if ls[0]=='Guava':
    sum+=1
  if ls[1]=='Kiwi':
    sum+=20
print(sum)
```

```
22
```

```
d=[[[1,2],[3,4]],[[5,6],[7,8]]]
def f(m):
  v=m[0][0]
  for x in m:
    for i in x:
      if v<i:
        v=i
  return(v)
print(f(d[0]))
```

```
4
```

```
a={(1,2):1,(2,3):2,(3,4):3} # We can use tuple as a key
print(a[3,4]) # And usko aise access kar sakte hai
```

```
3
```

```
a={'a':1,'b':2,'c':3}
print(a['a'])
```

```
1
```

```
a={}
def ad(i):
```

```
  if i in a:
    a[i]+=1
  else:
    a[i]=1001
ad('A')
ad('B')
ad('C')
ad('a')
ad('b')
ad('c')
print(len(a))
```

```
6
```

```
print(a)
```

```
{'A': 1001, 'B': 1001, 'C': 1001, 'a': 1001, 'b': 1001, 'c': 1001}
```

```
#
a={}
a[1]=1
a['1']=2
a[1]+=1

print(a)
```

```
{1: 2, '1': 2}
```

```
sum=0
for i in a: # i will always start from 0
  sum+=a[i]
print(sum)
```

```
4
```

```
d={}
d[1]=1
d['1']=2
d[1.0]=4 # 1.0 will replace 1
print(d)
sum=0
for i in d:
  sum+=d[i]
print(sum)
```

```
{1: 4, '1': 2}
6
```

```
m={}
m[(1,2,4)]=8
```

```
m[(4,2,1)]=10
m[(1,2)]=12
sum=0
for k in m:
  sum+=m[k]
print(sum)
print(m)
```

```
30
{(1, 2, 4): 8, (4, 2, 1): 10, (1, 2): 12}
```

```
box={}
jars={}
crates={}
box['biscuit']=1
box['cake']=3
jars['jam']=4
crates['box']=box
crates['jars']=jars
print(len(crates[box]))
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_1832/3035071434.py in <cell line: 0>()
      7 crates['box']=box
      8 crates['jars']=jars
----> 9 print(len(crates[box]))

TypeError: unhashable type: 'dict'
```

```
d={'c':97,'a':96,'b':98}
for i in sorted(d):
  print(d[i])
```

```
96
98
97
```

```
# id() gives the address of the variable
rec={'Name':'Python','Age':'20'}
r=rec.copy()
print(id(r)==id(rec)) # copy() does not mean moving
```

```
False
```

```
print(id(r),id(rec))
```

```
133516623164672 133516621744448
```

```
r={'a':123,'b':234}
id1=id(r)
del r
r={'a':123,'b':234}
id2=id(r)
print(id1==id2)
```

```
False
```

```
rn=[3,5,7,1,11,4,5,2]
for x in rn:
  if x==2 or x==4 or x==6 or x==8 or x==10:
    print("Even number found is",x)
```

```
Even number found is 4
Even number found is 2
```

```
for i in range(1,6):
  if(i!=3):
    print(i,' ',6-i)
```

```
1   5
2   4
4   2
5   1
```

```
for i,j in zip(range(1,6),range(5,0,-1)):
  if i==3 and j==3:
    continue

  print(i,' ',j)
```

```
1   5
2   4
4   2
5   1
```

File Handling in Python

Why file handling?

To share data over networks, read: r, write: w

```
# writing data to txt file
f=open("AP_02.txt",'w')
f.write("Amar\n")
f.write("Akbar\n")
f.write("Anthony\n")
print("Write operation done successfully")
```

```
Write operation done successfully
```

```
l=['A robin feathering its nest has very little time to rest, \n', 'collecting its bits of twigs and twine,\n','though quite intent in its pursuit, he has a merry tune
f.writelines(l)
print("Writing done")
f.close()
```

```
Writing done
```

```
# read operation
f=open("AP_02.txt",'r')
d=f.read()
print(d)
```

```
Amar
Akbar
Anthony
A robin feathering its nest has very little time to rest,
collecting its bits of twigs and twine,
though quite intent in its pursuit, he has a merry tune to toot
he knows a song can move the job along
```

```
# reading the number of characters
f=open("AP_02.txt",'r')
d=f.read(26)
print(d)
```

```
Amar
Akbar
Anthony
A robin
```

```
f=open("AP_02.txt",'r')
d=f.readlines()
print(d)
for x in d:
  print(x,end='')
f.close()
```

```
['Amar\n', 'Akbar\n', 'Anthony\n', 'A robin feathering its nest has very little time to rest, \n', 'collecting its bits of twigs and twine,\n', 'though quite intent in
Amar
Akbar
Anthony
A robin feathering its nest has very little time to rest,
collecting its bits of twigs and twine,
though quite intent in its pursuit, he has a merry tune to toot
he knows a song can move the job along
```

```
with open("AP_02.txt",'w') as f:
  f.write("Amar\n")
```

```
  f.write("Akbar\n")
  f.write("Anthony\n")
  print("Write operation done successfully", f.closed)

print("Is the file actually closed?", f.closed)
```

```
Write operation done successfully False
Is the file actually closed? True
```

```
# tell()
f=open("AP_02.txt",'r')
print(f.tell())
print(f.read(2))
print(f.tell())
print(f.read(4))
print(f.tell())
```

```
0
Am
2
ar
A
6
```

```
# r+ is the mode where it first reads, then writes, and vice versa with w+
da="This is what you give me to work with, but honey I've seen worse, \n I can turn this sow's ear into a silk purse"
f=open("AP_0201.txt",'w')
f.write(da)
with open("AP_0201.txt",'r+') as f:
  r=f.read()
  print(r)
  print("Current position of the cursor is:", f.tell())
  f.seek(111)
  print("Current position of the cursor is:", f.tell())
  f.write("....")
  f.seek(0)
  r=f.read()
  print(r)
```

```
This is what you give me to work with, but honey I've seen worse,
 I can turn this sow's ear into a silk purse
Current position of the cursor is: 111
Current position of the cursor is: 111
This is what you give me to work with, but honey I've seen worse,
 I can turn this sow's ear into a silk purse....
```

```
# WAP to print numbers of lines, words, characters present in the file

import os,sys
fname=input("Enter file name: ")
if os.path.isfile(fname):
  print("File exist:", fname)
```

```
  f=open(fname,'r')
else:
  print("File does not exist: ",fname)
  sys.exit(0)

lcount=wcount=ccount=0
for line in f:
  lcount=lcount+1
  ccount=ccount+len(line)
  words=line.split()
  wcount=wcount+len(words)

print("Total lines: ", lcount)
print("Total words: ", wcount)
print("Total characters: ", ccount)
```

```
Enter file name: AP_02.txt
File exist: AP_02.txt
Total lines:  9
Total words:  9
Total characters:  57
```